shugaba

FOR ERIK, DANIEL AND KHALID

Tech stack & design system

How Shugaba is built and how it should look. Read this before your first commit.

VERSION
1.0 · 29 Oct 2026

SOURCE OF TRUTH
docs/BUILD_SPEC.md

HACKATHON
PayPal AI Hackathon, Devpost

01 · THE PRODUCT

What we are building

Shugaba is the traditional rotating savings circle (adashi, ajo, esusu, susu) rebuilt so software holds the money, every member is guaranteed a full payout, and an AI organiser called Shugaba runs the circle in English, Pidgin and Hausa. *"The savings circle your grandmother trusted, with a guarantee that you get paid and a record that finally counts."*

No human custody
PayPal in, programmable treasury, automatic payouts.

Protection Pool
A 1–3% fee guarantees the full pot even if someone misses.

Shugaba
The AI organiser: reminders, plans, recovery, refusals.

Turn Market
Member agents negotiate turn swaps inside owner rules.

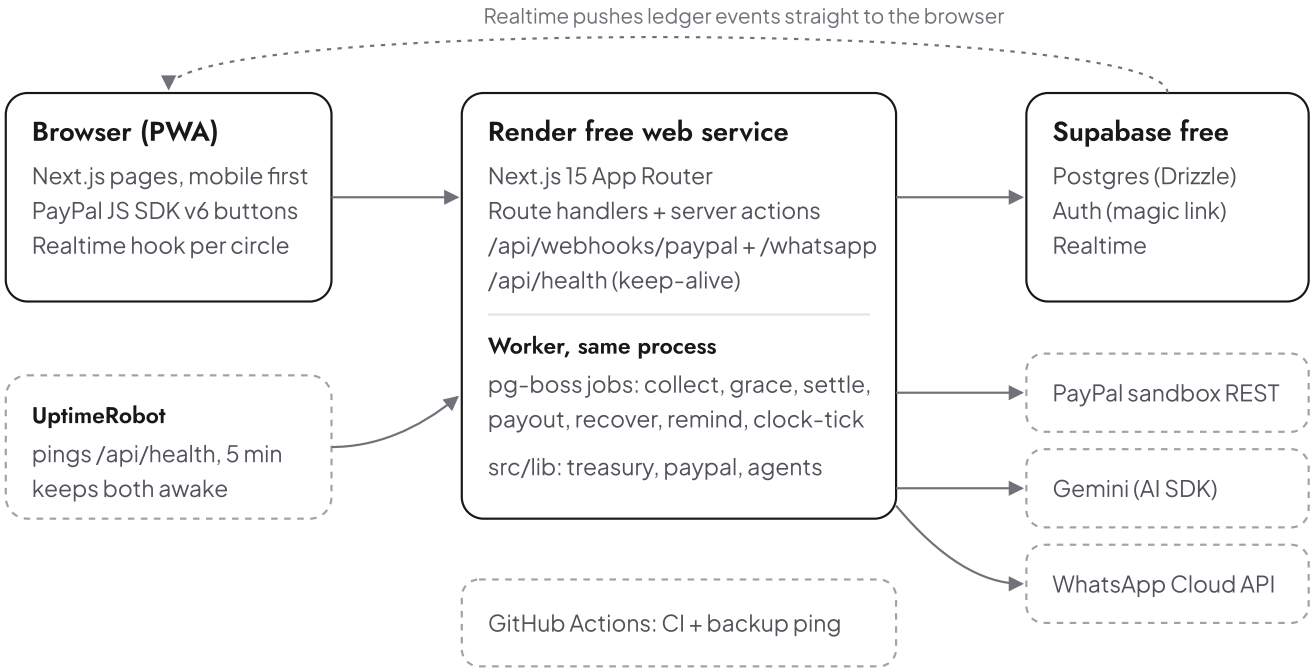
Circle Score
300–850, earned by paying on time. Shown as awards from Pebble.

The demo is the "golden path": Amina's 6-member Family Circle, $50 a round, $300 pot. Musa misses round 2, the pool covers him, Tunde swaps turns with Ifeoma for a $5 bonus, Musa pays back his invoice, Amina reaches Pillar. It is also our main end-to-end test.

02 · ARCHITECTURE

Two free services, one process

One Render web service runs Next.js and the job worker together. One Supabase project holds the database, auth and realtime. No Redis, no separate API, no monorepo tooling. Total hosting cost: $0.



Render free sleeps after about 15 minutes idle and Supabase free pauses after about 7 days idle. The UptimeRobot ping to `/api/health` (runs `select 1`) keeps both awake through judging.

Stack (locked)

Layer	Choice
Language	TypeScript, `strict: true`, one pnpm package (no monorepo)
Web + API	Next.js 15 App Router: pages, route handlers (REST + webhooks), server actions
UI	Tailwind CSS, shadcn/ui (restyled to our tokens), PWA manifest, `motion` only where CSS can't
Database	Supabase Postgres, free project
ORM	Drizzle ORM; migrations in git. RLS policies in `supabase/migrations`
Auth	Supabase Auth magic link. `DEMO_MODE` adds a “log in as” picker
Realtime	Supabase Realtime: Postgres changes + one broadcast channel per circle
Jobs + clock	pg-boss, started from `instrumentation.ts` when `RUN_WORKER=true`. Demo clock in `src/lib/treasury/clock.ts`
Payments	Own typed PayPal REST client in `src/lib/paypal` behind a `PaymentProvider` interface (PayPal + Mock). JS SDK v6 in the browser
AI	Vercel AI SDK. Gemini by default (`@ai-sdk/google` : Flash for Shugaba + recovery, Flash-Lite for fast tasks). Mock for tests
Notifications	Web Push + WhatsApp Cloud API (Meta direct, utility templates only, mock first)
Signatures	Node `crypto` Ed25519 for signing circle rules
Sponsor UI	Bryntum Scheduler (payout calendar), AG Studio (control tower). Both behind adapters with a plain fallback
i18n	next-intl. `en` complete; `ha` and `pcm` for core member screens
Tests	Vitest, fast-check (≥ 1,000 property runs on money logic), Playwright golden path
Quality	ESLint, Prettier, CI: lint, typecheck, tests, `pnpm audit`, gitleaks
Hosting	Render free web service (`render.yaml`) + Supabase free + UptimeRobot keep-alive. $0 total

Repo layout

```
shugaba/
  src/
    app/          routes, api/ (webhooks, circles, judge)
    components/   UI kit, payout-calendar/, control-tower/
    lib/
      protocol/   Zod schemas + shared types
      db/         Drizzle schema, client, seed
      treasury/   ledger hash chain, fees, settlement, clock (pure)
      paypal/     PaymentProvider: PayPalProvider + MockProvider
      agents/     llm.ts, Shugaba, member agents, recovery, evals/
      score/      Circle Score, levels, smart order
      realtime/   broadcast helper + useCircleEvents()
      theme/      tokens.ts, Tailwind preset
    worker/       pg-boss bootstrap + jobs/
  supabase/       config.toml, migrations (RLS, publication)
  drizzle/        generated migrations
  docs/           BUILD_SPEC.md, DECISIONS.md, design/
  e2e/            Playwright golden path
```

Run it locally

```
supabase start        # local Supabase
pnpm i
cp .env.example .env  # all mocks
pnpm db:migrate
pnpm db:seed
pnpm dev              # web + worker
# then open localhost:3000/judge
```

Scripts

`dev` `build` `start` `worker` `test` `test:e2e` `lint` `typecheck` `db:migrate` `db:seed` `db:reset` `eval:agents` `record:golden`

Local magic-link emails land in Inbucket at `http://127.0.0.1:54324`.

05 · SWITCHES

Modes and env

Everything real is switched on by env vars. The full list is in `.env.example`.

Variable	Values	Why
`PAYMENT_PROVIDER`	`mock` | `paypal`	Mock needs no keys. Sandbox via `PAYPAL_*`
`COLLECTION_MODE`	`vault` | `subscription` | `mock`	Vault lets our clock drive real sandbox charges
`LLM_PROVIDER`	`mock` | `google`	Only `src/lib/agents/llm.ts` imports a provider
`LLM_REPLAY`	`true` in demo	Golden-path agent outputs replay from fixtures. Saves Gemini quota
`DEMO_MODE`	`true` on the demo	Judge panel, log-in-as picker, fast clock. Refuses to boot with live PayPal
`WHATSAPP_MODE`	`mock` | `cloud`	Mock shows a WhatsApp-style preview on `/judge`
`CLOCK_SPEED`	1 to 1440	1440 = one day per minute

06 · RULES

Eight rules for everyone

1. **Money is an integer in cents.** Convert to decimal strings only inside `src/lib/paypal`. Any rounding remainder goes to the pool.
2. **Agents read and propose. Only the worker moves money.** And only after `src/lib/treasury` approves. No LLM output ever calls a money function.
3. **Ledger first.** Every state change writes a hash-chained `ledger_events` row before anything else.
4. **Main always runs with zero keys.** Mock PayPal, mock LLM, mock WhatsApp. Never block on a missing credential.
5. **Never guess PayPal behaviour.** Read developer.paypal.com and leave a `// VERIFY: <url>` comment on the call.
6. **Secrets live in `.env` and Render only.** Never in git, issues or chat. Service role key never `NEXT_PUBLIC_`.
7. **Validate everything with Zod.** Every route handler and server action parses input before logic.
8. **Small PRs to `main`.** Conventional Commits. CI must be green. Ambiguity goes in `docs/DECISIONS.md`.

Look and feel

White and greys, PayPal-clean, very few words. Shugaba (the Pebble) is the only brand colour. Green means one thing: money that has really arrived.

Do

- Hairlines (`--line`) to separate things
- Big Jost numbers, tabular figures for every amount
- One idea per screen, few words
- Black primary buttons, white text, 12px radius
- Touch targets ≥ 44px, usable at 390px wide
- Loading, empty, error, offline and “PayPal is down” states on every screen

Don’t

- Grey cards and pills everywhere (reads as generic AI)
- Coloured side borders or black underlines
- Green for anything except confirmed money
- “Live” dots, radar sweeps, mono labels, confetti
- Spinners on money
- Wordy explainers. Let Shugaba say one line

Colour tokens

In `src/lib/theme/tokens.ts` , exported as CSS variables and a Tailwind preset. Bryntum and AG Studio get the same theme.

–bg `#FFFFFF`
Page background

–surface `#F4F4F5`
Ask bar, sheets. Use sparingly

–line `#E4E4E7`
Hairlines and dividers

–text `#18181B`
Body text and primary buttons

–muted `#71717A`
Secondary text

–pebble `#3B358A`
Shugaba only. The one brand colour

–paid `#16A34A`
Confirmed money only

–accent `#0070E0`
PayPal blue: links, progress, thinking. Hover #003087

–grace `#B26A00`
Payment in grace / late

–covered `#1E63B5`
Covered by the Protection Pool

–failed `#C62828`
Failed payment, on hold

–gold `#B7860B`
Turn marker, swap and payout highlights

08 · TYPE

Typography

$300

Family Circle

Jost 500–700: headings, numbers, the wordmark. `font-variant-numeric: tabular-nums` on all amounts.

Got your $51, Tunde. Your part for round 3 is done.

Plus Jakarta Sans 400–700: everything else. Both from Google Fonts.

Use	Font	Size / weight
Hero amount	Jost	40–56px / 600
Screen title	Jost	24–28px / 600
Section label	Jost	13px / 500
Body	Plus Jakarta Sans	15–16px / 400
Small, captions	Plus Jakarta Sans	12–13px / 500, muted
Pay button label	“PayPal” bold italic	Real build uses the PayPal JS SDK button

09 · BRAND

Logo and Pebble

shugaba

Logo A: Pebble beside “shugaba” in Jost 700, lowercase. Pebble alone is the app icon. WhatsApp profile icons are in `adashi/brand/`.



Pebble is Shugaba. A flat indigo stone with two dot eyes, no mouth. He is on every screen and narrates every money moment. He breathes, blinks and glances; mood shows through a soft glow, never a face. Occasional reactions (reading glasses when checking, raised brows when surprised) are brief, not his permanent look.

watching
All good, default

thinking
Working on an answer

speaking
New message

guarding
Money is moving

pleased
Paid, done

concerned
Missed payment

alert
Needs attention

Avatars, states and levels

Default avatar is a grey person silhouette (`docs/design/avatars/`). Each avatar carries one small badge.

State	Badge
Paid	Green check
Gets paid today	Green plus, dotted money path from Pebble
Late	Amber clock
Covered	Blue shield
Paid out	Avatar faded
Left	Dashed outline, exit badge
Open seat	Dashed outline, grey plus
On hold	Red pause
PayPal not linked	Amber !
Loading	Shimmer

Circle Score levels: awards from Pebble

A small Pebble perches top-right on the avatar in the level colour, bigger at higher levels; Pillar glows. Internal enum stays `BRONZE / SILVER / GOLD / DIAMOND`.



Newcomer
300–549 · `#B7703F`



Steady
550–679 · `#8E949E`



Trusted
680–779 · `#C98E12`



Pillar
780–850 · `#6D5BD0`

Level-ups are Pebble speaking: *“From today I call you Trusted, Amina.”*

“Gets paid today”

A green dotted money path with a travelling green dot runs from Pebble to that member. Pebble looks at them. A tiny label shows the name and “$300 today” in green. The avatar itself stays plain.

Motion

Calm and alive, never busy. Screens ease in (fade + 8px rise, 220ms, ease-out); lists stagger 30ms. Money numbers count up once (600ms). Buttons press to 0.97. Status changes cross-fade colour. Everything is instant under `prefers-reduced-motion`.

Shugaba’s voice

First person, max three short sentences, every money claim has an amount, a date and a ledger link. Greets in the member’s language. Never hype, never shames a defaulter in front of the group, says “I can’t do that” plainly.

Example: “Musa missed this round, so the Protection Pool covered his $50. Ngozi got the full $300 on time.”

Routes

Route	Screen
`/`	Landing, “Try the demo”
`/login`	Magic link; demo picker
`/home`	My circles, score, next payment
`/circles/new`	Create circle wizard
`/join/[code]`	Rules, sign, connect PayPal
`/c/[id]`	Circle home: ring around Pebble
`/c/[id]/calendar`	Payout calendar (Bryntum)
`/c/[id]/feed`	Shugaba’s timeline of money events
`/c/[id]/ledger`	Ledger + verify chain

Route	Screen
`/c/[id]/chat`	Full chat with confirm cards
`/c/[id]/trades/[id]`	Agent negotiation + confirm
`/c/[id]/rules`	Numbered rules with anchors
`/score`	Score, level, events
`/notifications`	From Shugaba
`/settings`	PayPal, language, WhatsApp, leave
`/verify/[id]`	Public verifier (initials only)
`/judge`	Judge panel
`/ops`	Control tower (operators)

Home layout: a ring of member avatars around Pebble, Shugaba’s one current line under it, and an ask bar at the bottom (text + mic; voice is stretch).

Where the designs are

What	Where
Screen exports	`docs/design/screens/` : one PNG per screen, in flow order
Pebble	`docs/design/shugaba-pebble.html` (states), `shugaba-pebble-age.html` (reactions)
3D Pebble prototype	`docs/design/shugaba-3d.html`
Logo and emblem	`docs/design/logo-concepts-v2.png` , `shugaba-emblem.html`
Avatars	`docs/design/avatars/member-1…6.svg`
App icons	`adashi/brand/` : Pebble icon, indigo and white, SVG + PNG
Source files	Project folder `adashi/design/` and `adashi/brand/` ; copy into the repo as `docs/design/`

The build spec (`SHUGABA_BUILD_SPEC.md`) may lag the newest design changes. Where the screen exports and the spec disagree on looks, follow the screens. Where they disagree on money rules, ask Erik.